Elasticity

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Demand

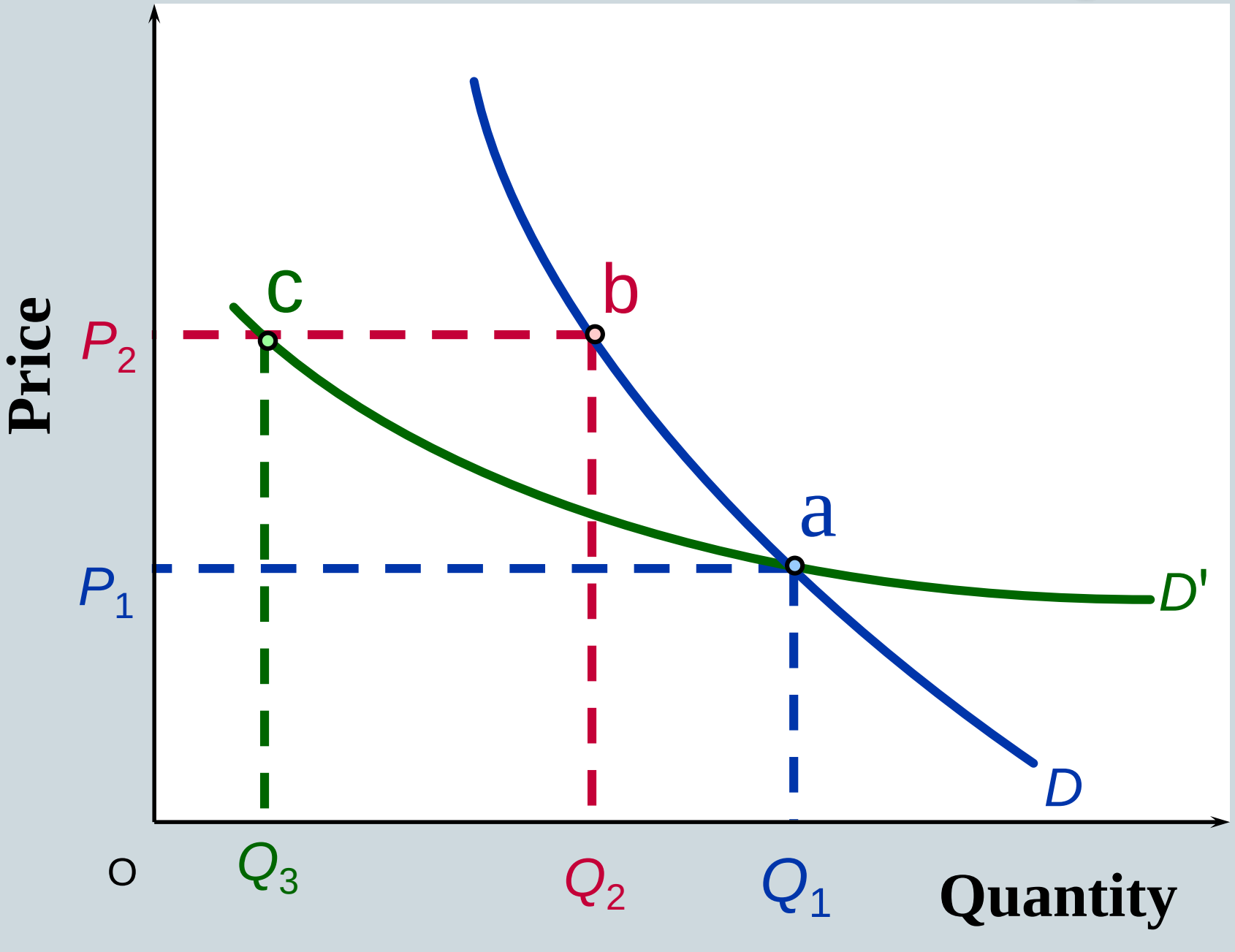
- The market we are going to examine is highly competitive with many competing firms and many consumers.

  - ✓ This means that firms and consumers are price takers

Introduction

- As pointed out earlier, when the price of a good rises (or falls), quantity demanded falls (rises).
- Economists would like to know by how much quantity demanded falls or rises in response to a price change
- In other words, we would like to know how responsive demand is to price changes.
- For instance, consumers' response to a change in the price of oil would differ from that of Voltic Mineral Water.

Market demand and Price Change



Elasticity

- **Elasticity** is the economic measure of the response of one variable to a change in another
- **Elasticity of demand** measures the degree of responsiveness of quantity demanded to changes in the determinants of demand.
- Since not all the factors that affect demand can be measured quantitatively, we will discuss three types of demand elasticity:
  - **Price Elasticity of Demand**
  - **Income Elasticity of Demand and**
  - **Cross-Price Elasticity of demand**

Price Elasticity of demand

- Price elasticity of demand measures the degree of responsiveness of quantity demanded to changes in the commodity's own price.

- Elasticity compares the size of the change in quantity demanded to that price.

- Since quantity and price are measured in different units, the only sensible way to measure elasticity is to use proportionate or percentage changes.

Measurement of Price Elasticity:

$$Price\ Elasticity\ of\ Demand$$

$$= \frac{\%\boldsymbol{Change\ in\ Quantity\ Demanded}}{\%\ \boldsymbol{Change\ in\ Price}}$$

Or

$$\boldsymbol{PED} = \frac{\%\Delta \boldsymbol{Q_D}}{\%\ \Delta \boldsymbol{P}}$$

Measurement of Price Elasticity:

Percentage change in quantity is given as:

$$\frac{Q_1 - Q_0}{Q_0} \times 100\% \quad \text{------(1)}$$

Percentage change in Price is given as:

$$\frac{P_1 - P_0}{P_0} \times 100\% \quad \text{-------(2)}$$

Price Elasticity of demand

- Dividing equation (1) by equation (2) implies:

- $$PED = \frac{\frac{Q_1 - Q_0}{Q_0} \times 100\%}{\frac{P_1 - P_0}{P_0} \times 100\%}$$
- Solving this will yield an alternative method.
- This is given as:

$$PED = \frac{Q_1 - Q_0}{P_1 - P_0} \times \frac{P_0}{Q_0}$$

Price Elasticity of demand: Example

- Assume that the price of Yam decreases from GH¢5 to GH¢4 per tuber and this causes the quantity demanded to increase from 5 tubers per day to 10 tubers. What is the price elasticity of demand?

- From this information,

$$P_0 = 5\ , \quad P_1 = 4, \quad Q_0 = 5 \quad Q_1 = 10$$

Price Elasticity of demand: Example

$$PED = \frac{10-5}{4-5} \times \frac{5}{5}$$

$$= \frac{5}{-1} \times 1$$

$$= \frac{5}{-1}$$

$$= -5$$

Price Elasticity of demand: Example

Percentage change in quantity is given as:

$$\frac{10-5}{5} \times 100\% = 100\%$$

Percentage change in Price is given as:

$$\frac{4-5}{5} \times 100\% = -20\%$$

$$PED = \frac{\%\Delta Q_D}{\%\,\Delta P} = \frac{100\%}{-20\%} = -5$$

Interpretation of Elasticity Figures

- Since demand curves are generally downward sloping, it implies that percentage change in price and that of quantity would have opposite signs.
- That is, a percentage increase in price would be accompanied by a percentage decrease in price and vice versa.
- Either way, the price elasticity of demand is always negative**. *In this class, I would ignore the negative sign and interpret the absolute values.***

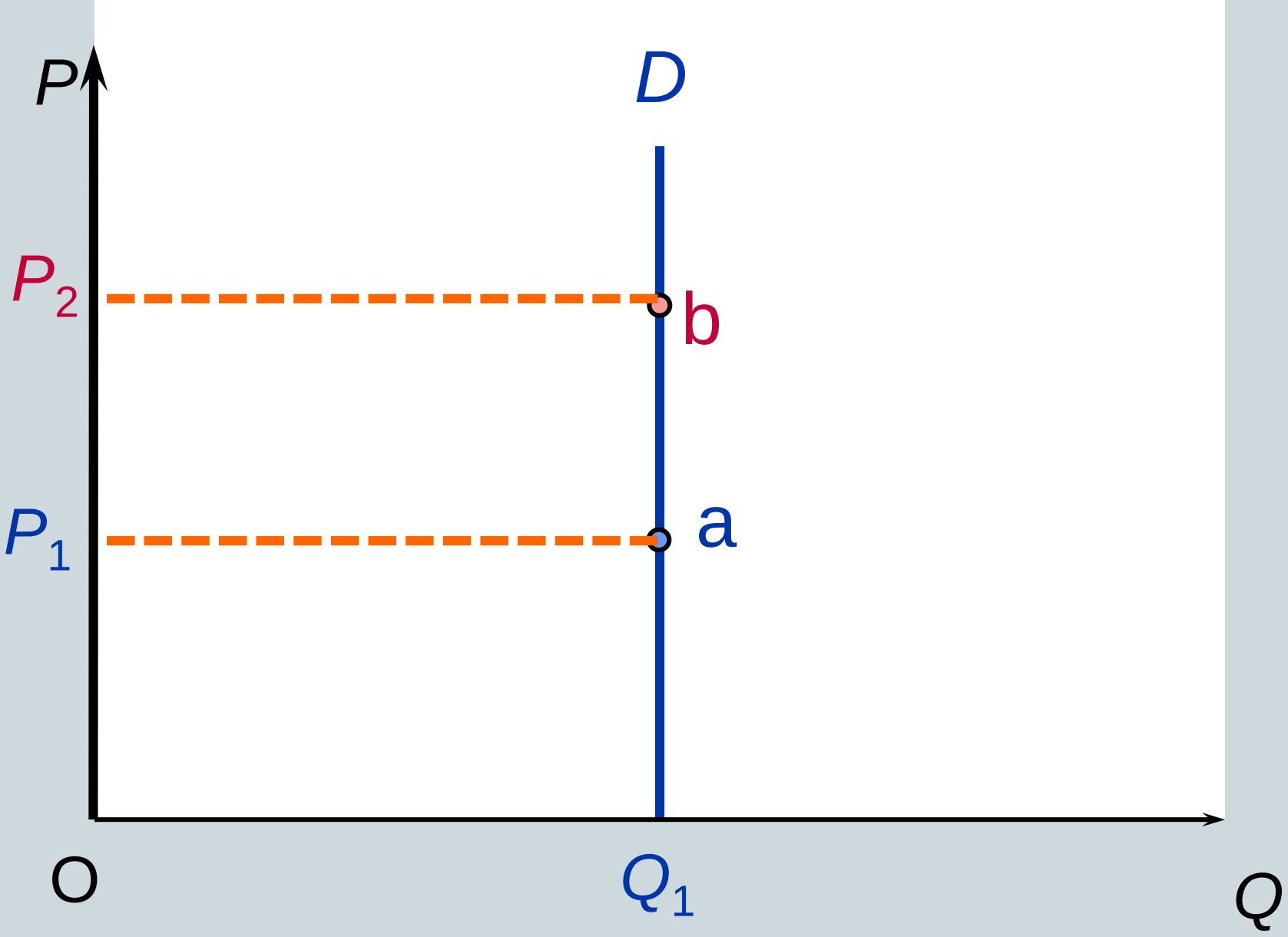
Interpretation of Elasticity Figures

- ***If PED = 0, then demand is Perfectly inelastic.***

- This means that demand does not respond at all to changes in price. That is, no matter the price, quantity demanded is the same.

- The demand curve is thus vertical as shown below:

Perfectly inelastic demand ($P\hat{I}_D = 0$)



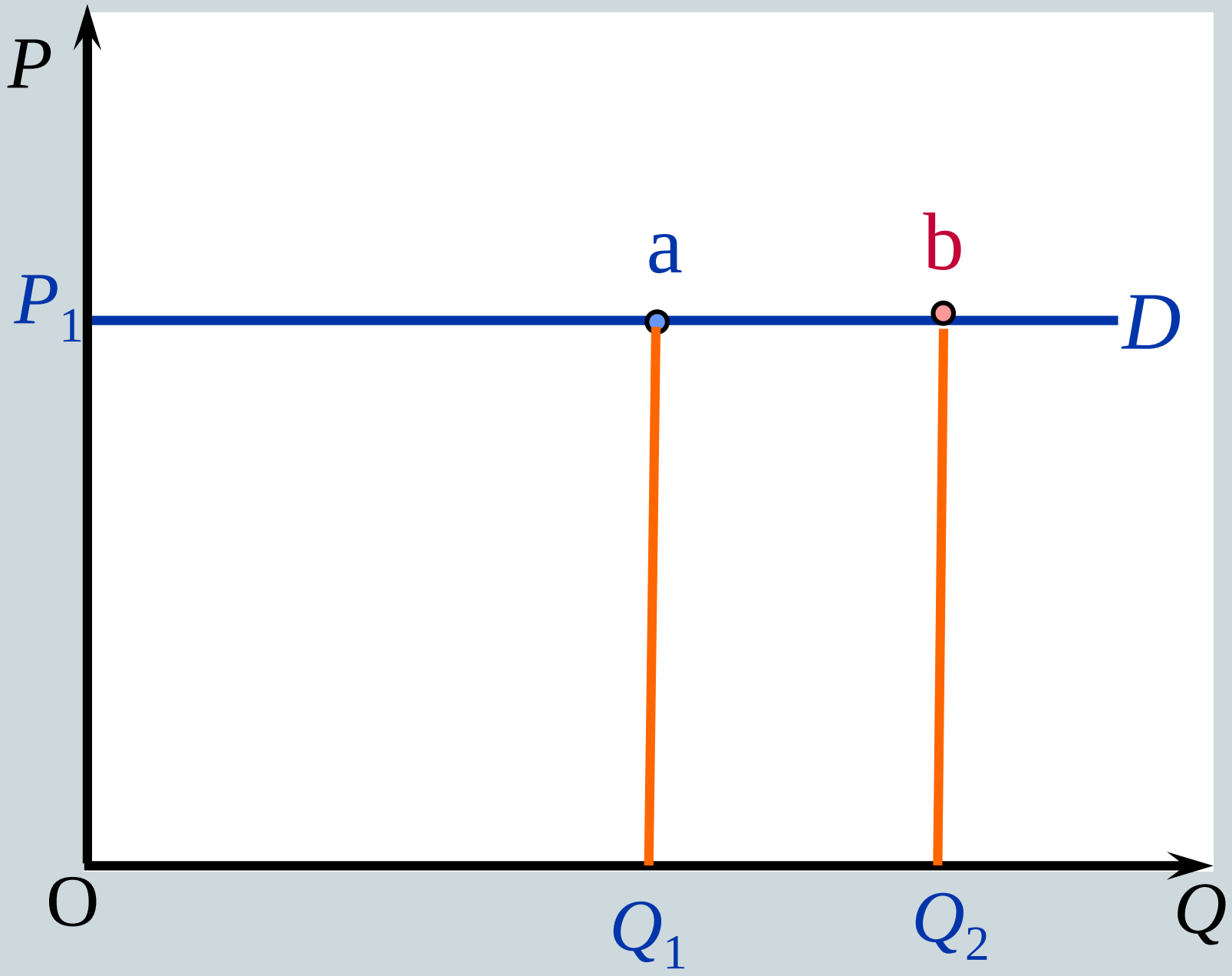
Interpretation of Elasticity Figures

- ***If PED = ∞, then demand is Perfectly Elastic.***

- This means that demand responds to a small change in price. That is, a small change in price brings about infinitely large change in quantity demanded.

- This is shown by a horizontal demand curve

Infinitely elastic demand ($P\hat{I}_D = ¥$)

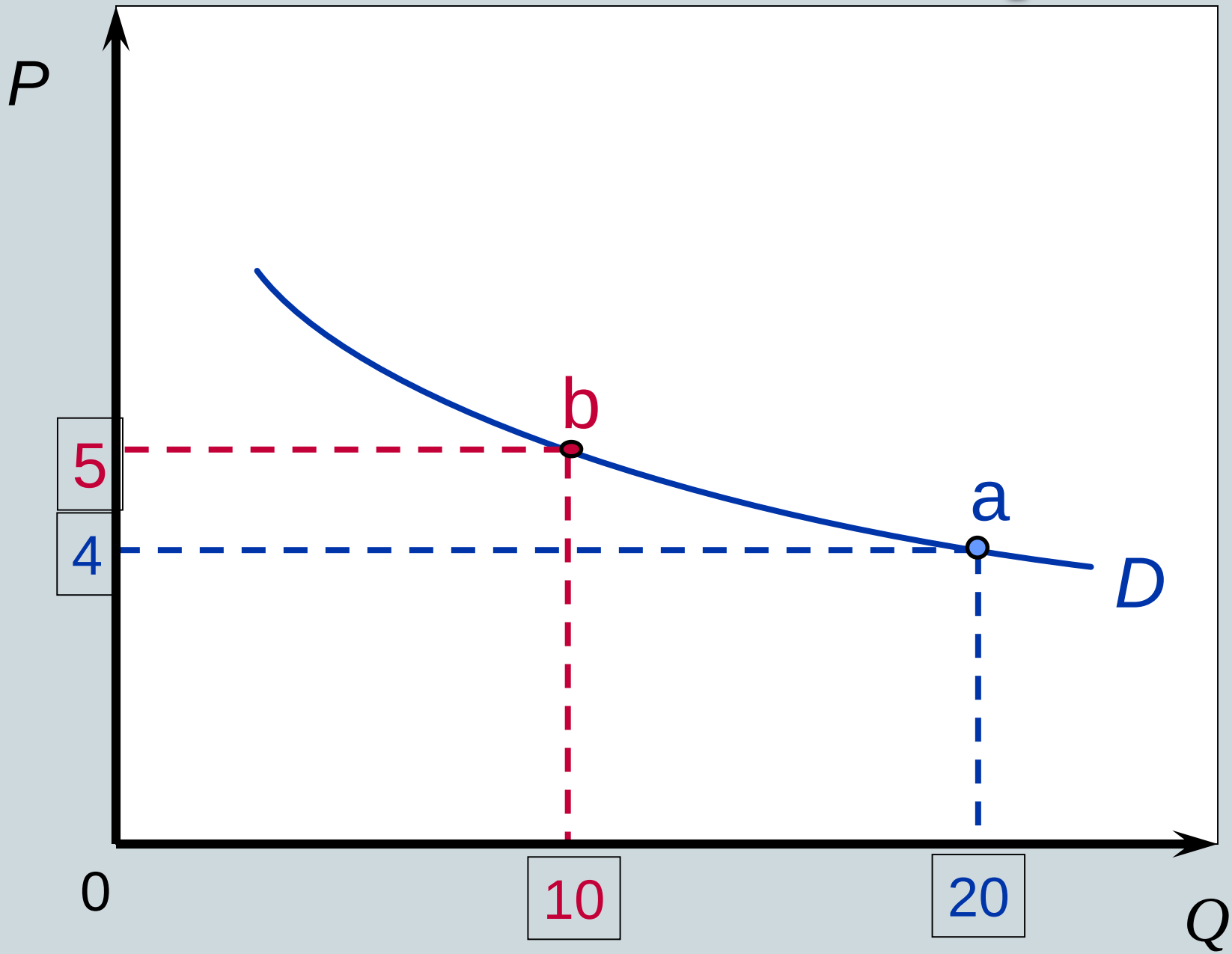


Interpretation of Elasticity Figures

- ***If PED>1, then demand is Elastic.***
- This is where a change in price causes proportionately larger change in the quantity demanded. In this case, the value of elasticity will be greater than 1 since we are dividing a larger figure by a smaller figure.

- This is shown by a relatively flatter demand curve

Elastic demand between two points

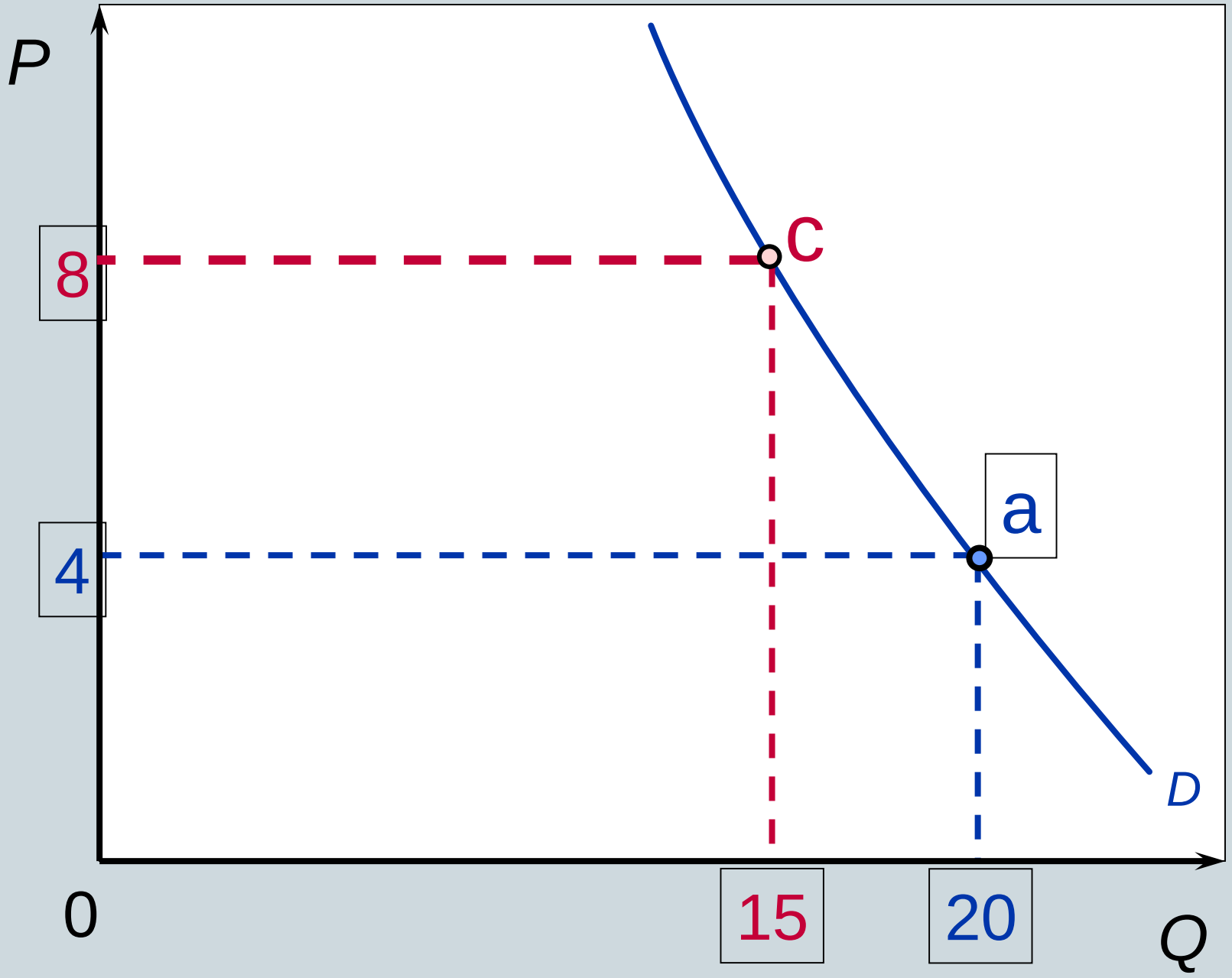


Interpretation of Elasticity Figures

- ***If 0<PED<1, then demand is Inelastic.***
- This is where a change in price causes proportionately smaller change in the quantity demanded. In this case, the value of elasticity will be less than 1 since we are dividing a smaller figure by a larger figure.

- This is shown by a relatively steeper demand curve

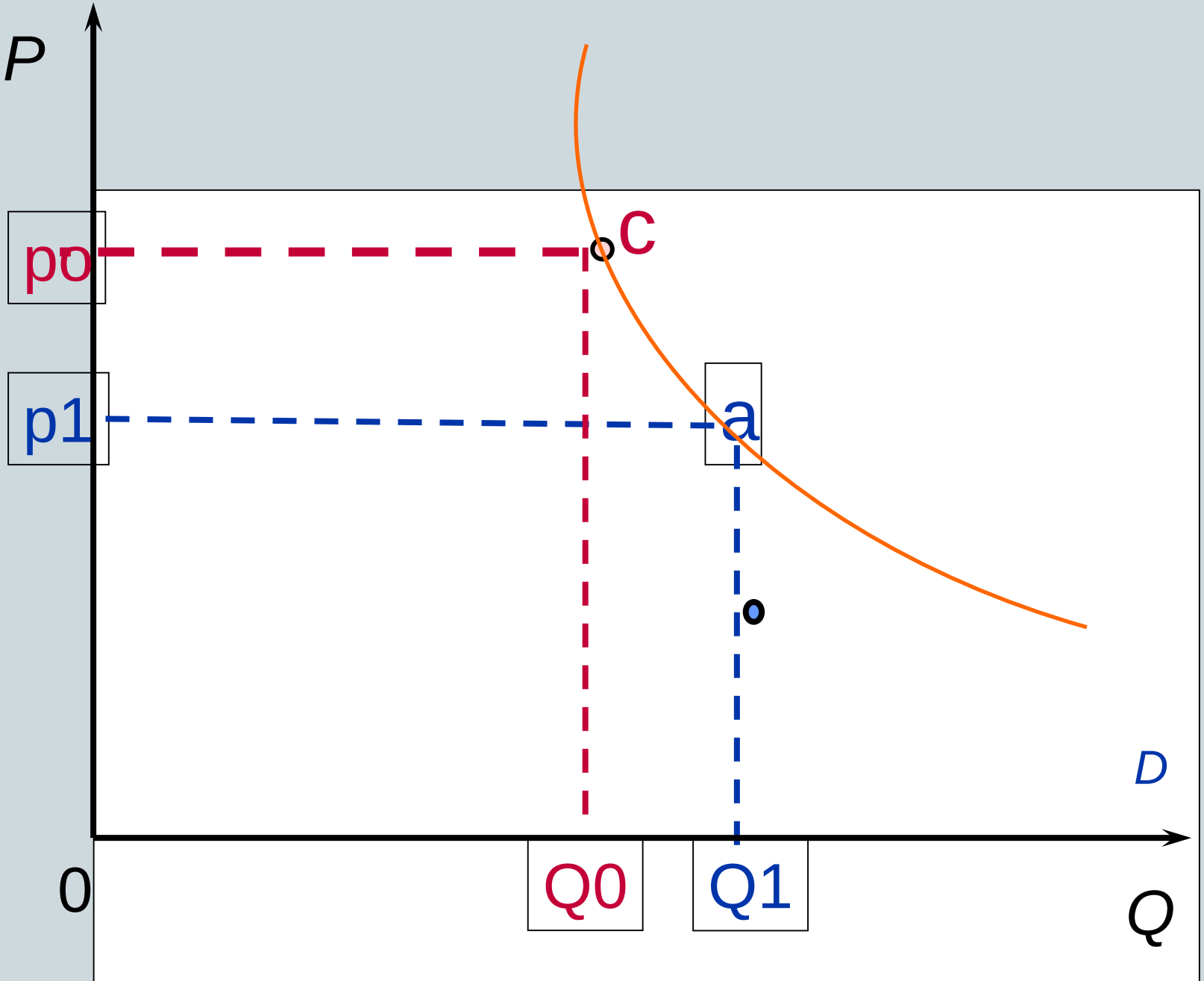
Inelastic demand between two points



Interpretation of Elasticity Figures

- ***If PED =1, then demand is said to be unitary Elastic.***
- This is where price and quantity demanded change by the same proportion. In this case, the value of elasticity will be 1 since we are dividing a figure by itself.
- The demand curve is a rectangular hyperbola:

Unitary Elastic



Summary of Price Elasticity of Demand

- Elastic Demand - Ed will be > 1
- Inelastic Demand - Ed will be < 1
- Unitary Elastic Demand - Ed will be = 1
- Perfectly Inelastic Demand - Ed will be = 0
- Perfectly Elastic Demand - Ed will be = ∞

Determinants of price elasticity of demand

- number and closeness of substitute goods
- Number of uses of the commodity
- the proportion of income spent on the good
- The number of new buyers
- time

Price elasticity of demand and consumer expenditure

- ***Total Revenue (expenditure) = P x Q***

- A price increase has two effects on revenue:
  - **Higher *P* means <u>more revenue</u>** on each unit you sell.
  - **But you sell <u>fewer units</u> (lower *Q*),** due to Law of Demand.
- Which of these two effects is bigger?
- It depends on the price elasticity of demand.

Price elasticity of demand and revenue

- If demand is elastic, then

price elasticity of demand > 1

% change in ***Q*** > % change in ***P***

- ***The fall in revenue from lower Q is greater than the increase in revenue from higher P, so revenue falls.***

Price elasticity of demand and revenue

- If demand is **inelastic**, then

price elasticity of demand < 1

% change in Q $<$ % change in P

- ***The fall in revenue from lower Q is smaller than the increase in revenue from higher P, so revenue rises.***

Price elasticity of demand and revenue

- If demand is **Unitary elastic**, then

price elasticity of demand = 1

% change in Q = % change in P

- ***The fall in revenue from lower Q is the same as the increase in revenue from higher P, so revenue remains the same.***

Summary of Price Elasticity of Demand (E_d) & r/ship with total revenue

If we take the the absolute value of the price elasticity of demand (i.e. ignore the sign) $|E_d|$ we can conclude:

Where:

- the absolute value of E_d is greater than 1: $|E_d| > 1$
- Demand is price ELASTIC

- the absolute value of E_d is less than 1: $|E_d| < 1$
- Demand is price INELASTIC

$|E_d$ (Apples)$|$ = 2 $\Rightarrow$ **ELASTIC** → ↑ P ⇒ ↓ TR

$|E_d$ (Petrol)$|$ = 0.2 $\Rightarrow$ **INELASTIC** → ↑ P ⇒ ↑ TR

Price elasticity of demand and revenue

	Price Rise	Price Decline
Elastic ($E_D > 1$)	TR decreases	TR increases
Unit Elastic ($E_D = 1$)	TR constant	TR constant
Inelastic ($E_D < 1$)	TR increases	TR decreases

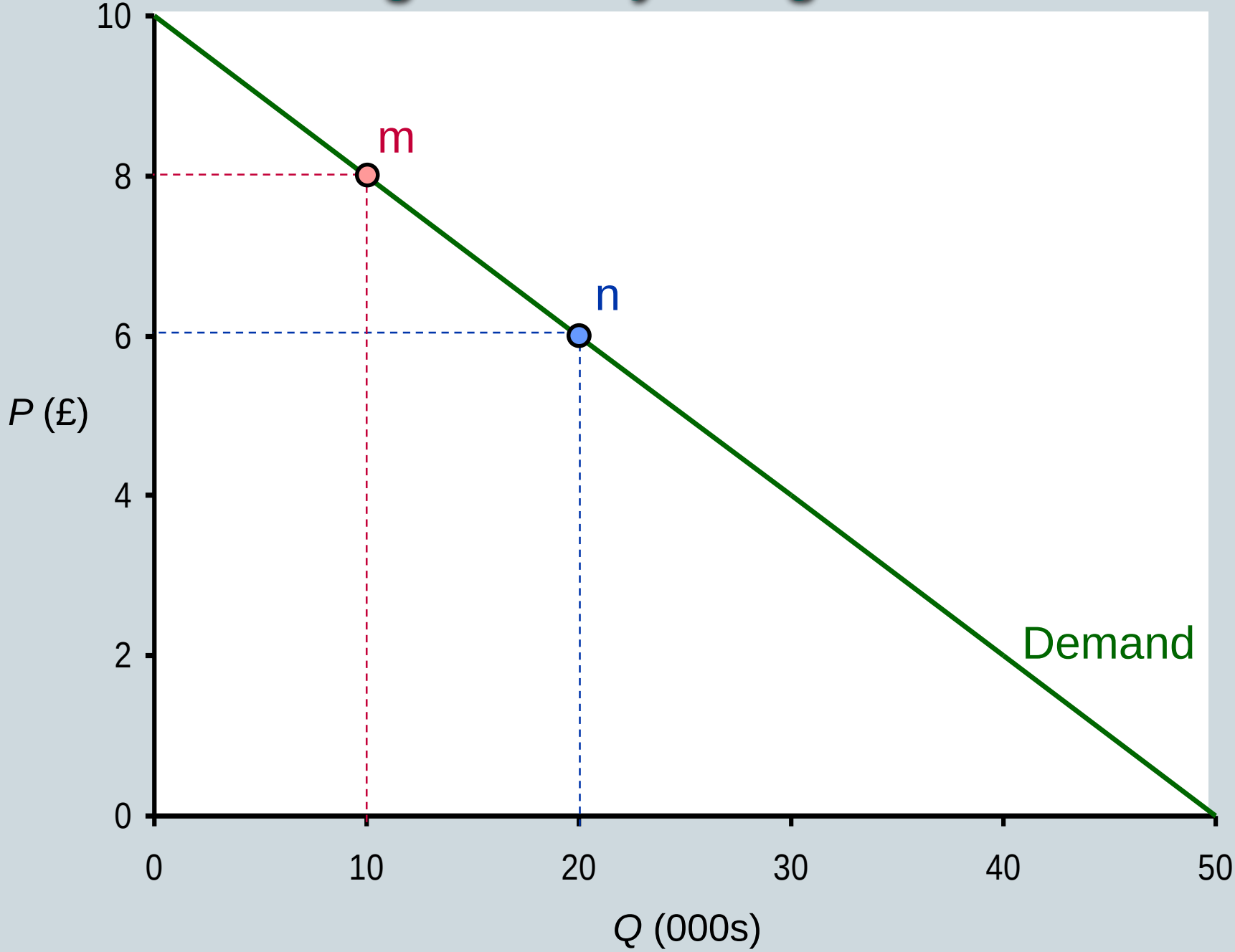
Arc Elasticity

- Arc elasticity measures elasticity between two point on the demand curve.
- We $\Delta Q/averageQ \div \Delta P/average\ P$
- *Or*

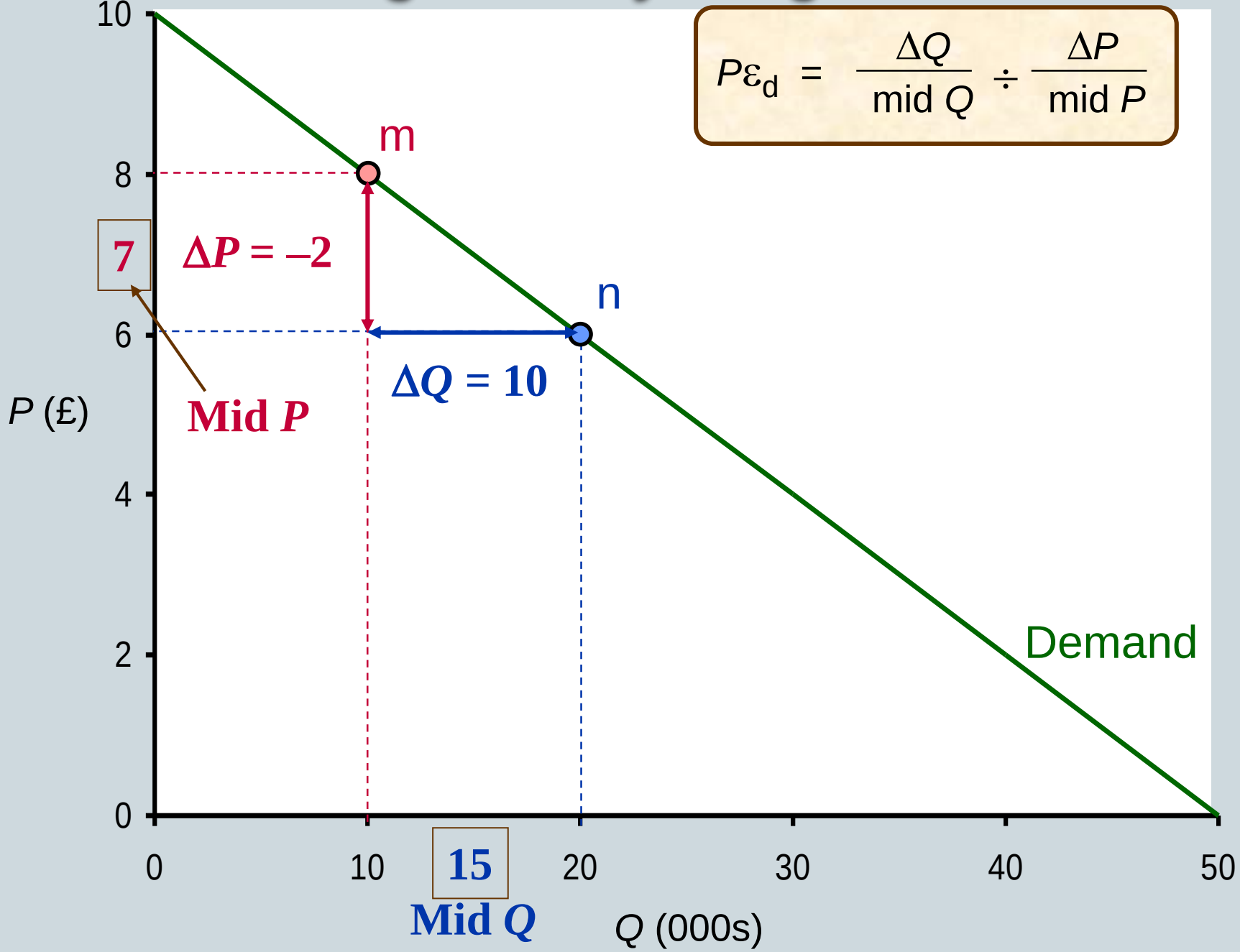
$$Arc\ Price\ elasticity\ of\ demand = \frac{\Delta Q}{\Delta P} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$$

$$Arc\ Price\ elasticity\ of\ demand = \frac{(Q_2 - Q_1)}{(P_2 - P_1)} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$$

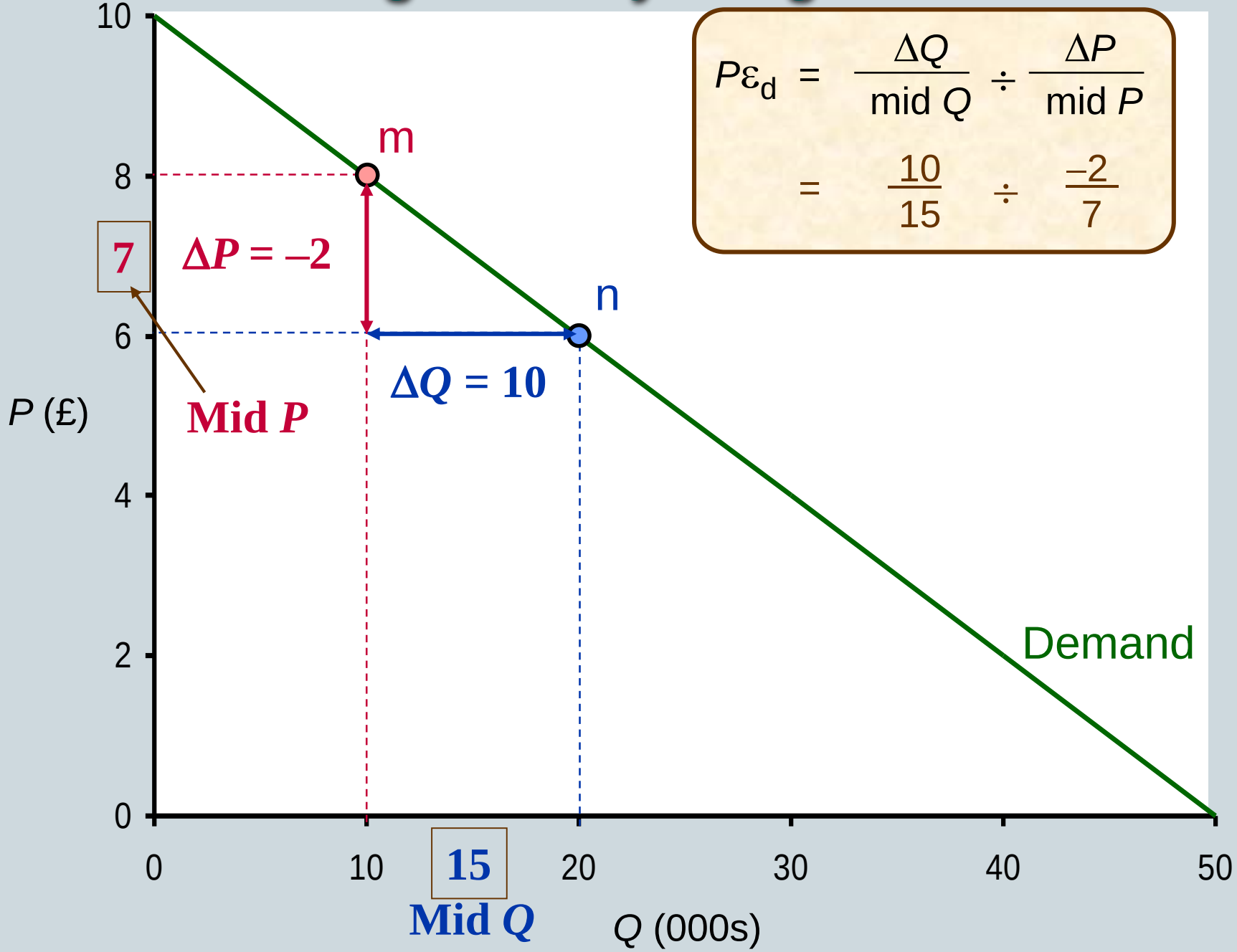
Measuring elasticity using the arc method



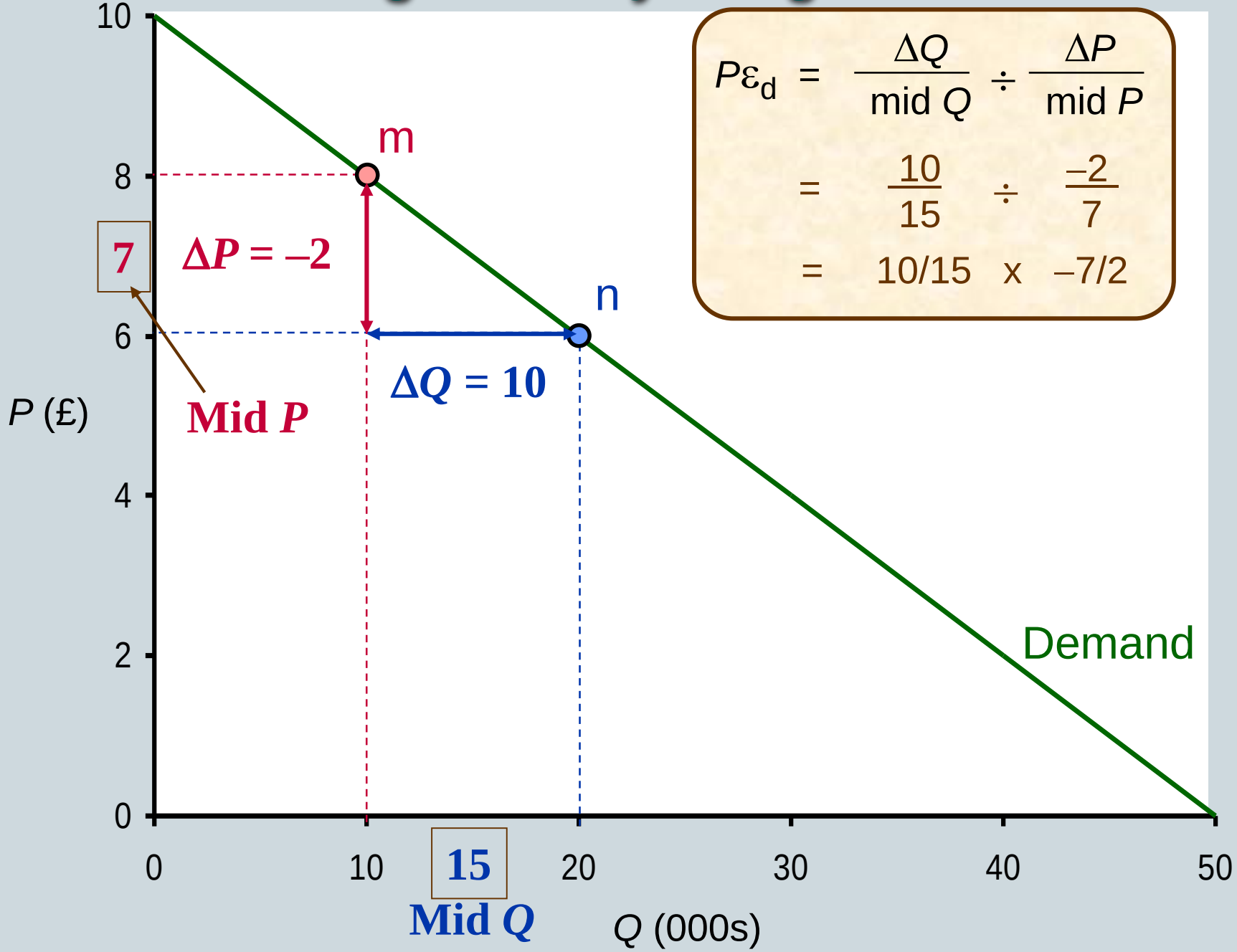
Measuring elasticity using the arc method



Measuring elasticity using the arc method



Measuring elasticity using the arc method

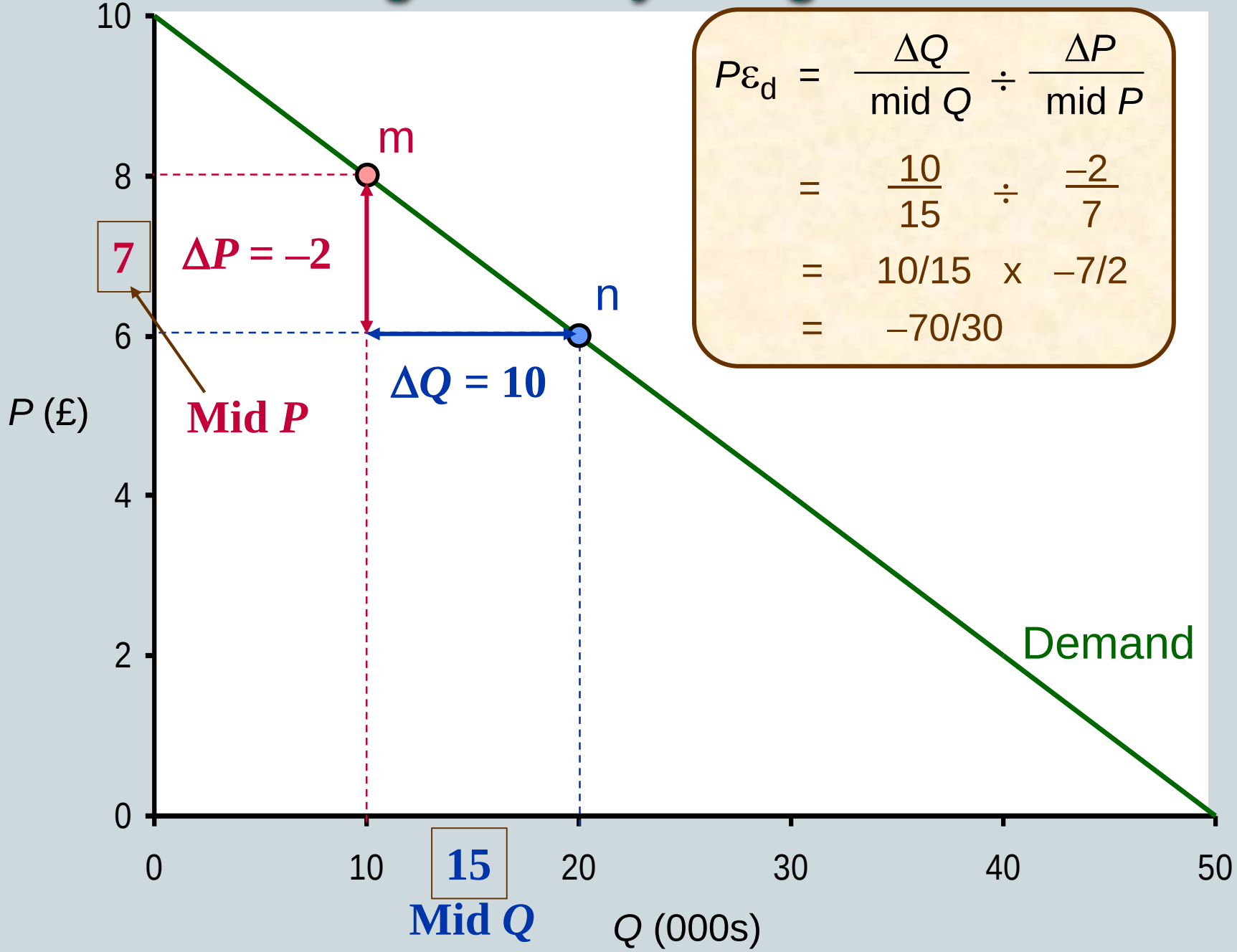


$$P\varepsilon_d = \frac{\Delta Q}{\text{mid } Q} \div \frac{\Delta P}{\text{mid } P}$$

$$= \frac{10}{15} \div \frac{-2}{7}$$

$$= 10/15 \times -7/2$$

Measuring elasticity using the arc method



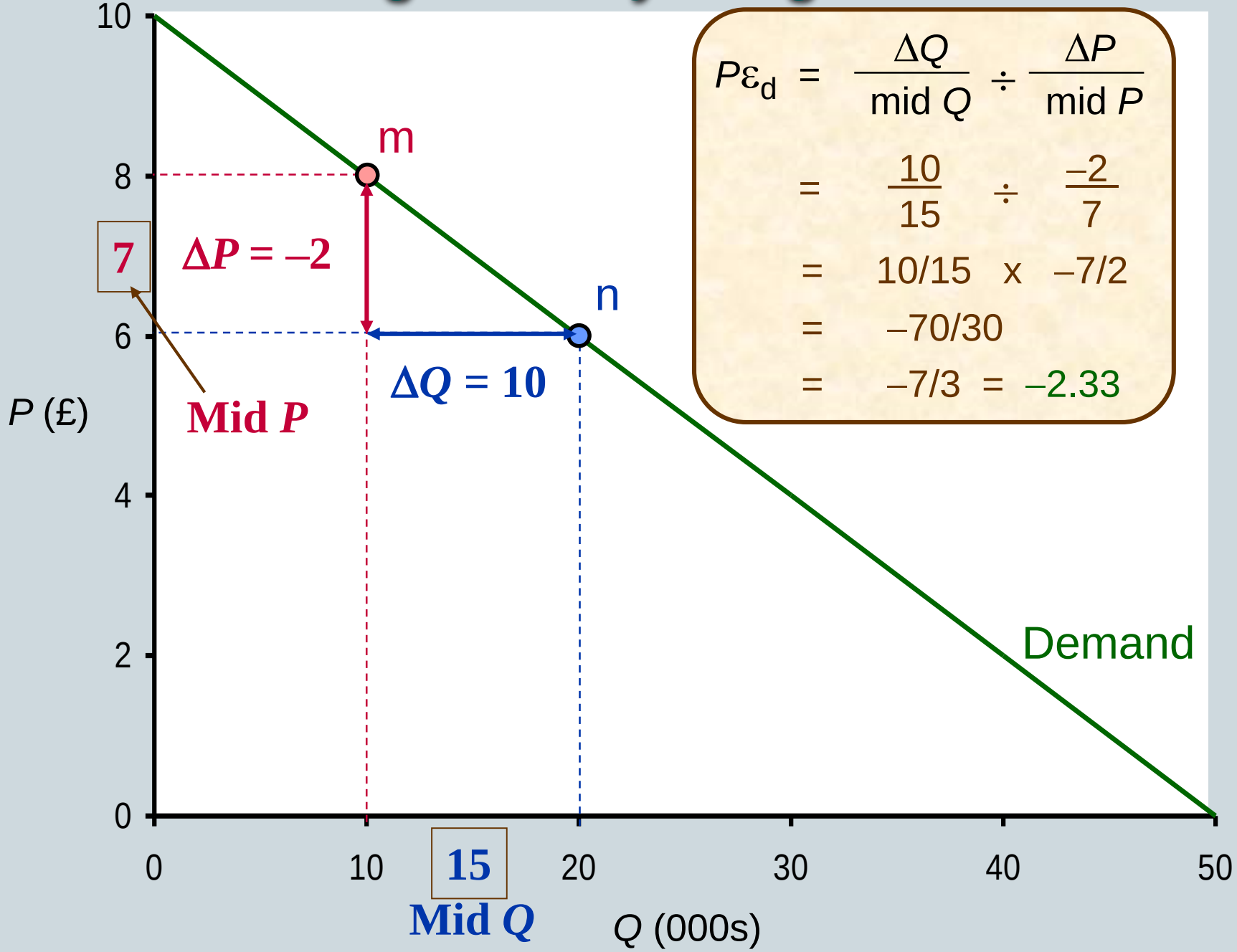
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$$= -70/30$$

Measuring elasticity using the arc method



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$$= \frac{10}{15} \div \frac{-2}{7}$$

$$= 10/15 \times -7/2$$

$$= -70/30$$

$$= -7/3 = -2.33$$

Q If the price of good X rises from £9 to £11 and as a result quantity demanded falls from 100 units to 60 units, what is the price elasticity of demand between these prices?

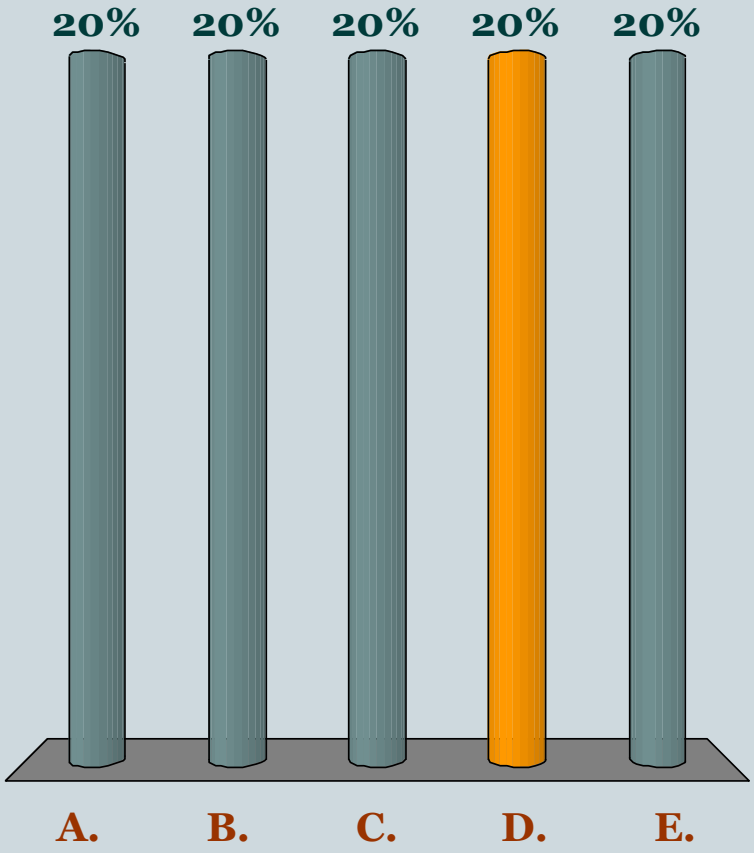
A. 2/−80 = −0.025

B. −80/2 = −40

C. 0.2/−0.5 = −0.4

D. −0.5/0.2 = −2.5

E. −1



Check your knowledge (cont'd)

Q3. When demand is price inelastic, what is the relationship between price and total revenue?

a. They move in the same direction.

b. They move in opposite directions.

c. When price changes, total revenue remains the same.

d. They are unrelated.

Check your knowledge (cont'd)

Q3. When demand is price inelastic, what is the relationship between price and total revenue?

a. They move in the same direction.

b. They move in opposite directions.

c. When price changes, total revenue remains the same.

d. They are unrelated.

Check your knowledge (cont'd)

Q4. Fill in the blanks: If an increase in the price of a substitute (Product B) leads to _____ in quantity demanded (Product A), the cross-price elasticity of demand is _____ .

a. an increase; positive

b. an increase; negative

c. a decrease; positive

d. a decrease; negative

Check your knowledge (cont'd)

Q5. Which of the following is likely to be most price inelastic in supply in the short run(i.e. it cannot be quickly changed in size)?

a. A baker.

b. A cattle farm.

c. A pizza shop.

d. A clothing retailer.

Check your knowledge (cont'd)

Q5. Which of the following is likely to be most price inelastic in supply in the short run?

a. A baker.
b. A cattle farm.
c. A pizza shop.
d. A clothing retailer.

Note the cattle farm cannot be expanded quickly if the price of cattle rise. For the other producers it is quicker for them to increase supply if their prices increase.

Check your knowledge (cont'd)

Q5. A local store has a store-wide 10% off sale (e.g., price falls by 10%). They sell 20 percent more quantity of shirts, but only 5 percent more quantity of belts.

a. What are the price elasticities of demand for shirts?

b. What are the price elasticities of demand for belts?

c. Comment on the results obtained in 'a' and 'b'.

Shirts: $\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{+20\%}{-10\%} = -2$ … $usually$: $\left|\frac{\%\Delta Q}{\%\Delta P}\right| = |-2| = 2$

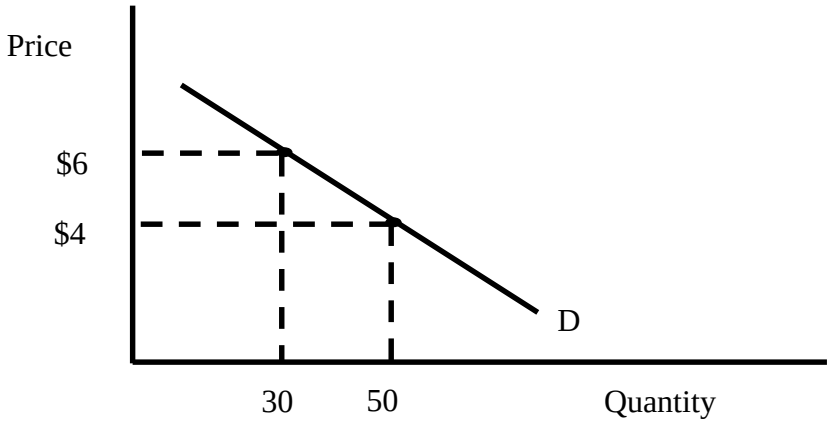
Belts: $\varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{+5\%}{-10\%} = -\frac{1}{2} = -0.5$ … $usually$: $\left|\frac{\%\Delta Q}{\%\Delta P}\right| = |-0.5| = 0.5$

Consumer purchases of shirts are **MORE** responsive to the change in price…they are MORE elastic. That is why we use absolute values, 2 > 0.5 [but -2 < -0.5].

Check your knowledge (cont'd)

Q7. Using the information below, calculate the price elasticity of demand arc approach and the.

Price	Quantity
$4	50
$6	30



We know the absolute change in price (|ΔP|) is $2 [$6-$4=$2] and the absolute change in quantity (|ΔQ|) is 20 [50-30=20]

but what do we put in the denominators?

We use the **MIDPOINTS** (or averages) of the two values ($5 for price; 40 for quantity)

$$\text{So: } \%\Delta Q = \frac{\Delta Q}{\text{midpoint Q}} = \frac{20}{40} = \frac{1}{2} = 0.5 = 50\%$$

$$\text{So: } \%\Delta P = \frac{\Delta P}{\text{midpoint P}} = \frac{2}{5} = 0.4 = 40\%$$

$$\text{So: } \varepsilon = \frac{\%\Delta Q}{\%\Delta P} = \frac{50}{40} = \frac{5}{4} = 1.25$$

Check your knowledge (cont'd)

Q8. Explain how a microeconomic principle or concept you learned in MME291can be applied to a real-world problem.

Q9. What is the difference between disagreeing about a positive statement and disagreeing about a normative statement?

Check your knowledge (cont'd)

Q10. Sketch a Production Possibility Frontier curve showing Product A and Product B on the respective axes. On your sketch demonstrate how this model shows increasing opportunity costs as more of Product A has to be given up to produce more of Product B. (You can use arrows or made-up numbers to show how more of A has to be given up to obtain more of B.)

b)On your PPF sketch above, use three points labelled i, ii, & iii, to demonstrate:

i) an inefficient combination of Products A & B;

ii) an efficient combination of Products A & B; and

iii) a currently unobtainable combination of Products A & B.

Check your knowledge (cont'd)

Q10. The supply curve in Figure 1 shows the monthly market for wool jumpers at a local craft market. For each of the following events, draw the new outcome for the supply curve.

a)The price of wool increases (i.e. an input cost).

b) Demand for jumpers decreases.

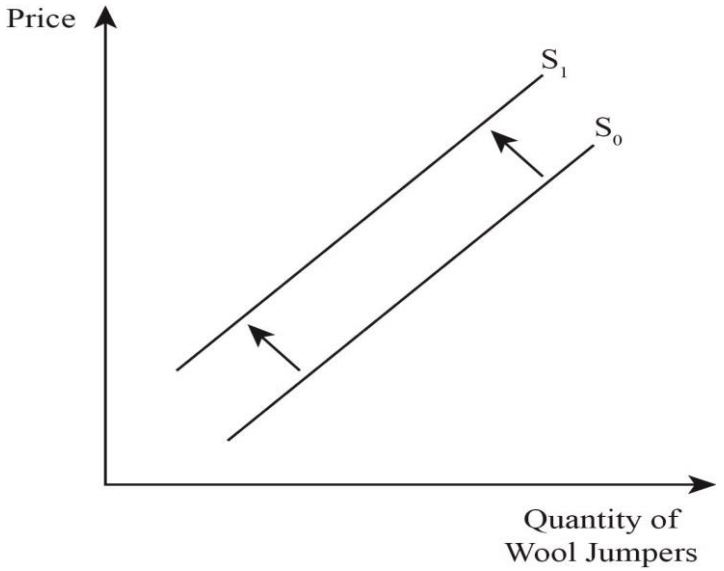
c) The price of knitting machines decreases (i.e. an input cost).

d) Consumers' incomes increase which increases the demand for jumpers.

Check your knowledge (cont'd)

a)The price of wool increases (i.e. an input cost).

- **A change in the price of input is a change in a non-price determinant and the <u>supply curve shifts left</u> from S_0 to S_1**



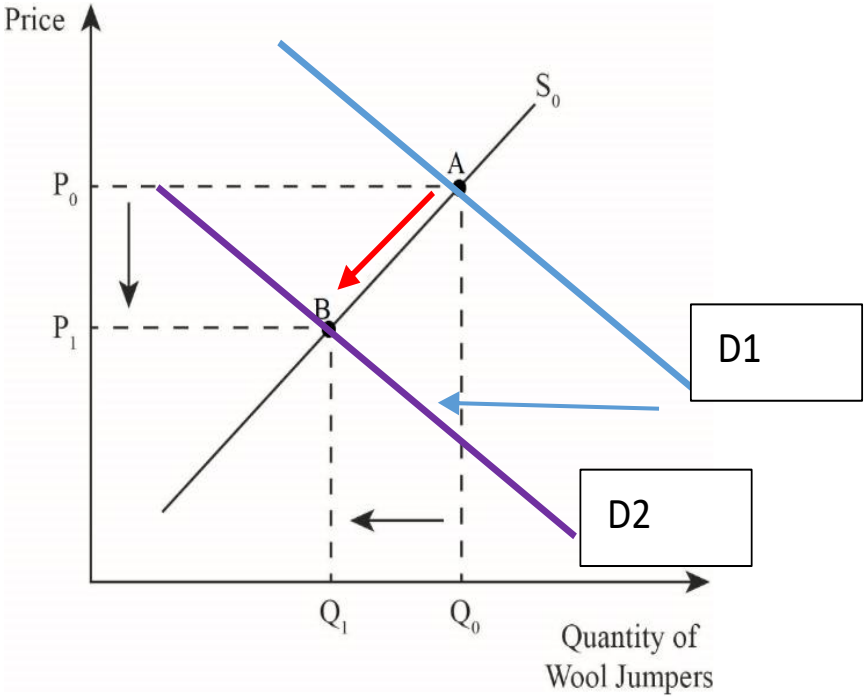
Check your knowledge (cont'd)

b) Demand for jumpers decreases.

- **A decrease in the demand for jumpers will decrease the price of jumpers, causing a <u>movement (downward) along the supply curve</u> from A to B.**
- **Explanation: This <u>represents a change in quantity supplied</u>, not a change of supply [i.e. the supply curve itself does not shift in this case]. The movement along the supply curve to a lower equilibrium price is due to the entire demand curve shifting to the left (i.e. downward), representing a decrease in demand at each and every price.**

Check your knowledge (cont'd)

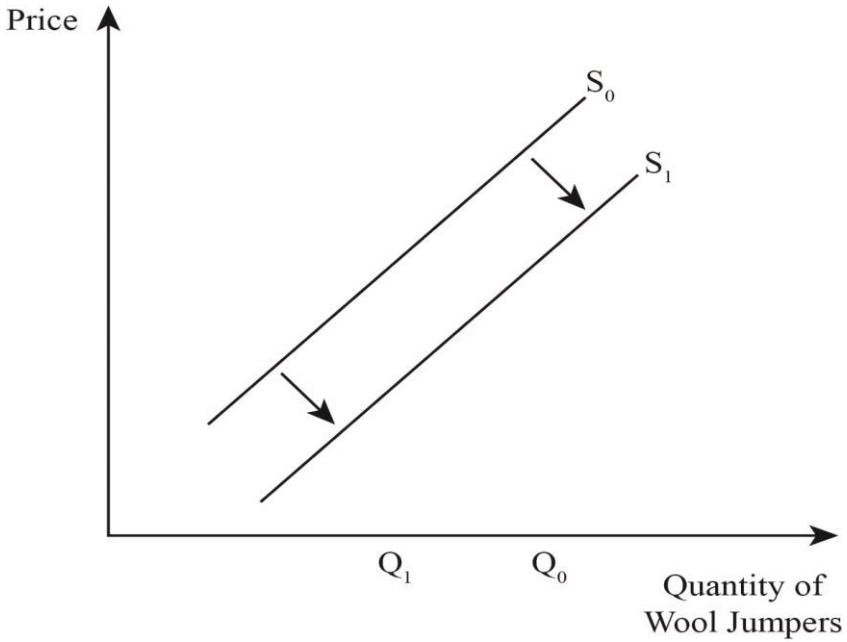
b) Demand for jumpers decreases.



Check your knowledge (cont'd)

c) The price of knitting machines decreases (i.e. an input cost).

Technological change causes supply to shift rightwards (*ceteris paribus*)



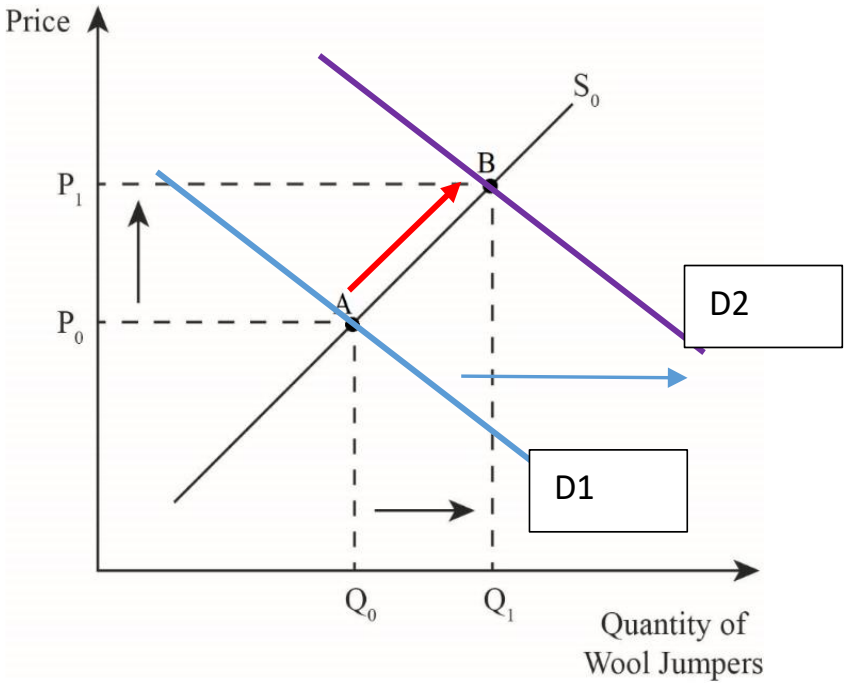
Check your knowledge (cont'd)

d) Consumers' incomes increase which increases the demand for jumpers.

- **An increase in the demand for jumpers will increase the price of jumpers, causing movement (upward) along the supply curve from A to B.**
- **Explanation: This is a change in quantity supplied and not a change of supply. It is the demand curve that shifts to the right (i.e. upwards) in this case. The shift of the entire demand is because consumers can purchase more units at each and every price because their incomes are higher. The shift of the demand curve to a higher equilibrium price is a movement along the supply curve.)**
-

Check your knowledge (cont'd)

d) Consumers' incomes increase which increases the demand for jumpers.



Check your knowledge (cont'd)

12) Using Demand and Supply analysis, demonstrate what policies could be enacted to make housing more affordable in Ghanaian cities. Use several sentences to accompany your diagram to explain what is happening.

Check your knowledge (cont'd)

12) Using Demand and Supply analysis, demonstrate what policies could be enacted to make housing more affordable in Ghanaian cities. Use several sentences to accompany your diagram to explain what is happening.

- **In this case it is <u>easier for governments to increase supply</u> (and lower prices, *ceteris paribus*), than to try and decrease demand.**
- **(*An acceptable answer is drawing a D & S diagram showing the Supply of houses increasing [shifting to the right or downwards] because of government policy. Also add a few sentences explaining why this is the easiest option compared with trying to shift D to the left.)***

Check your knowledge (cont'd)

- **Sources of Supply-side increases:**
- **Governments could increase housing stock by making more land available for residential building (e.g. using former industrial land) ; increase public housing, etc. (***This first point probably easiest with respect to policy);*

- **Elimination of stamp duty which adds to transaction and hence supply costs.**

- **Lower housing construction costs due to new technologies;**

Check your knowledge (cont'd)

- **Sources of Demand-side decreases** *(note that decreasing demand in the hope of decreasing housing prices is much more challenging to address compared with addressing the supply-side)***:**

❖ **More controlled immigration, especially in major cities** *(but controversial and hard to implement as governments in a democracy cannot tell people where they should live!)***;**

❖ **Abolishment of capital gains tax incentives which have caused housing to become an investment instrument rather than its traditional purpose of providing shelter) (***but recent election showed this is politically unpopular as the party that was proposing this lost***) ;**

❖ **Rising interest rates which would make loans more expensive and hence decrease demand, *ceteris paribus (****But this is a Central Bank decision and they have to consider the overall macroeconomy and not just one sector such as housing).*